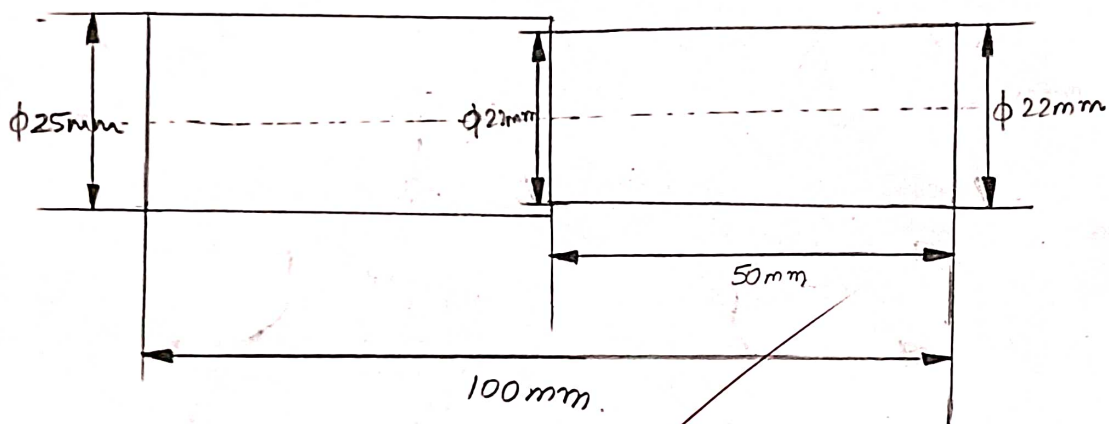


Machine

Shop



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FULL NAME..SHRUTI...GUPTA.....

CLASS..1st...Batch..F.T...

SHOP..MACHINE.....SHOP.....

DATE.....

## PRACTICE-1

Date of starting - 27/2/24

Date of finishing - 27/2/24

Objective :- Study of lathe machine and to make a job by facing, plane turning, step turning and chamfering operations as per given drawing.

Materials Required :- MS Rod - 25mm x 100mm

Tools and Equipments Used :-

1) Measuring And Marking Tool

(a) Marking block

(b) Engineering Scale

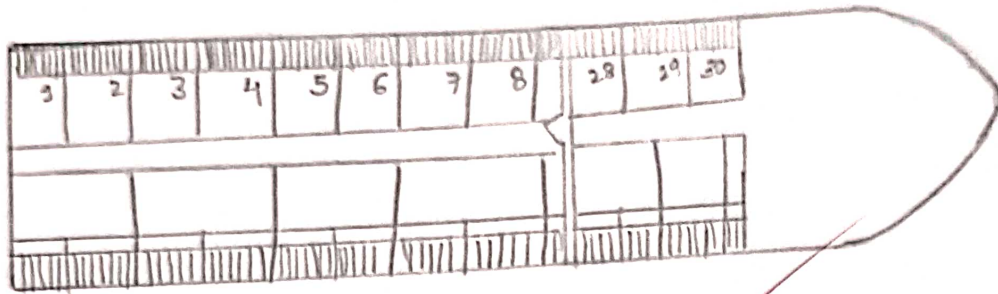
(c) Vernier Caliper

(d) Outside Caliper

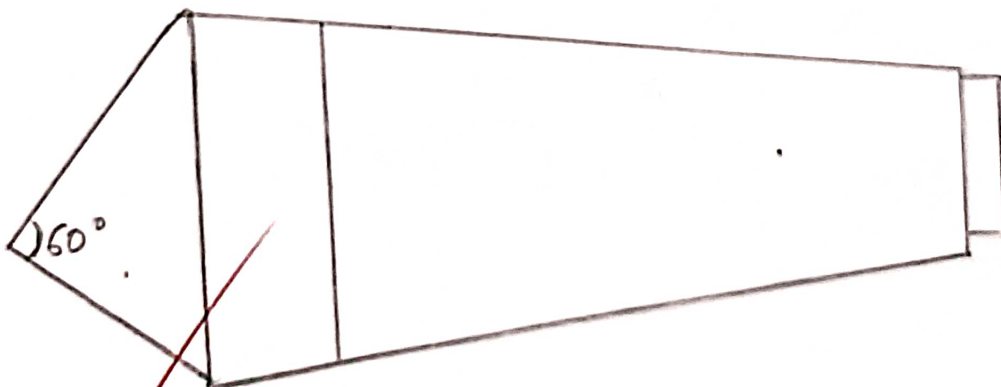
2) Holding And Supporting Tool

(a) Chuck - (1) Three jaw chuck

(2) Four jaw chuck



Engineering Scale.



Dead centre

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(b) Centre - (1) Dead centre  
(2) Live centre

(c) Tool post

3> Cutting Tools - (1) Turning tool  
(2) Chamfering tool.

4> Lathe machine

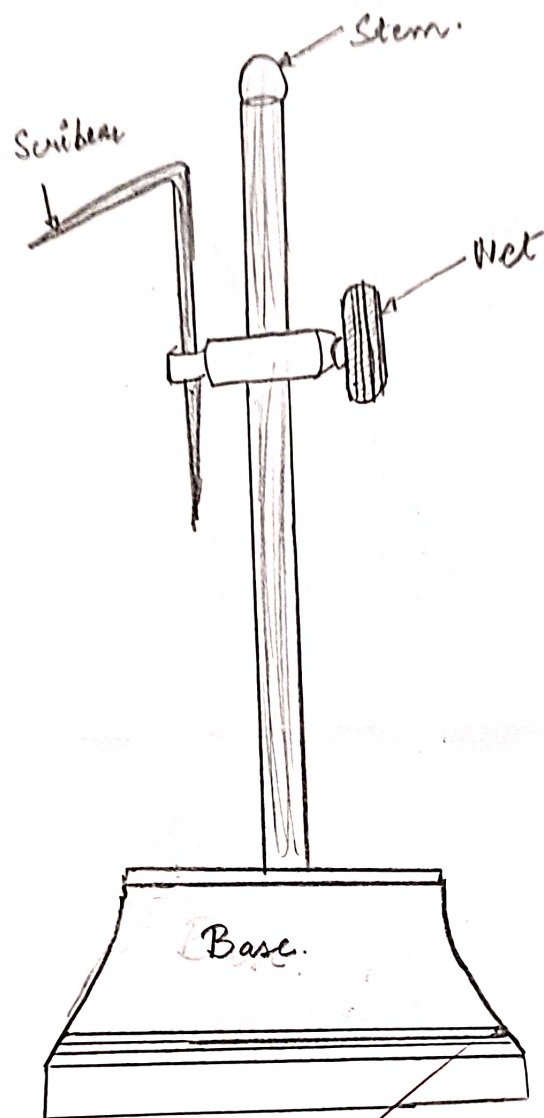
5> Power hacksaw machine

### Description of Tools And Equipments :

1> Marking Block :- It consists of a cast iron, sliding base fitted with a vertical steel rod called stem. The marker is fitted into an adjustable device carrying a knurled nut the scriber can be loosened or tightened to set it at desired position. It is mainly used during centering of job along lathe axis more precisely.

2> Engineering Scale :- It is a stiff straight steel strip having all the faces machine true. On one of the flat faces graduations are marked in inches and centimetres. The least count is  $\frac{1}{64}$  inch and 0.5 mm.

3> Outside Callipers :- These devices are used for making measuring & transferring the outside dimension.



Marking Block.

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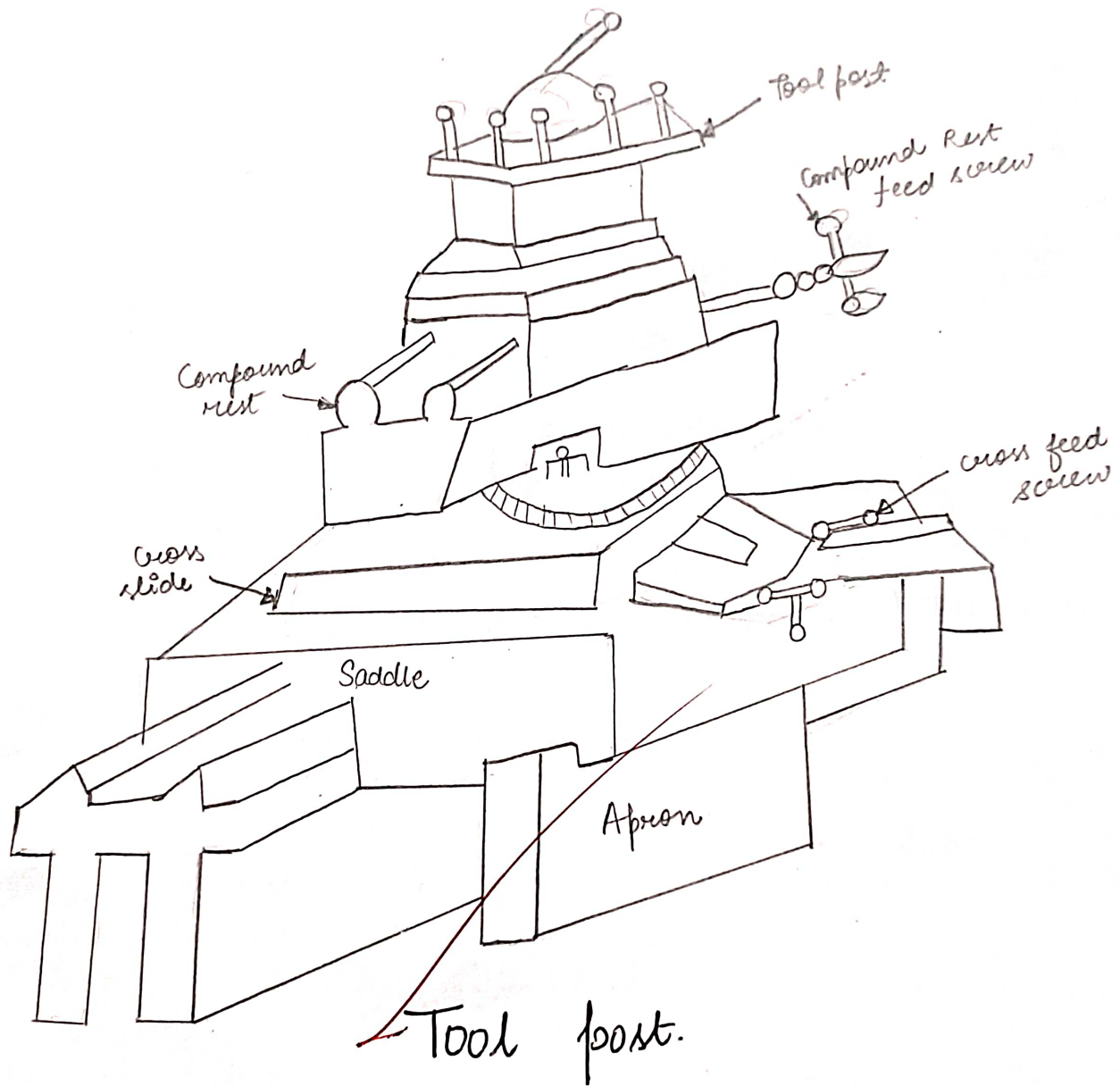
of components as shown in figure.

2. (a) Chuck :- They provide very efficient & true device for holding the work on the lathe during the operation. There are fairly large variety of chucks used on lathe.

i) Three jaw chuck :- It is also known as self-centering chuck. It consists of a cylindrical body having three jaws fixed radially at its front. It carries through hole at its centre to enable the long job to project backwards in the spindle. The chuck is directly mounted on the spindle.

ii) Four - jaw chuck :- It is also known as independent chuck. The rear position of its jaws is threaded & is engaged with separate adjusting screws with the result all the jaws can be moved separately & adjusted at a designed distance from the centre of the chuck. Thus enables the chuck to successfully hold irregular or eccentric jobs.

b) i) Dead centre :- The most common method of holding the workpiece in lathe is between the two centres. They are made of very hard material. It is fitted in tail stock & remains stationary. This centre due to high friction proves it to be inefficient at high



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speeds. Centres are made of cast steel or high grade tool steel.

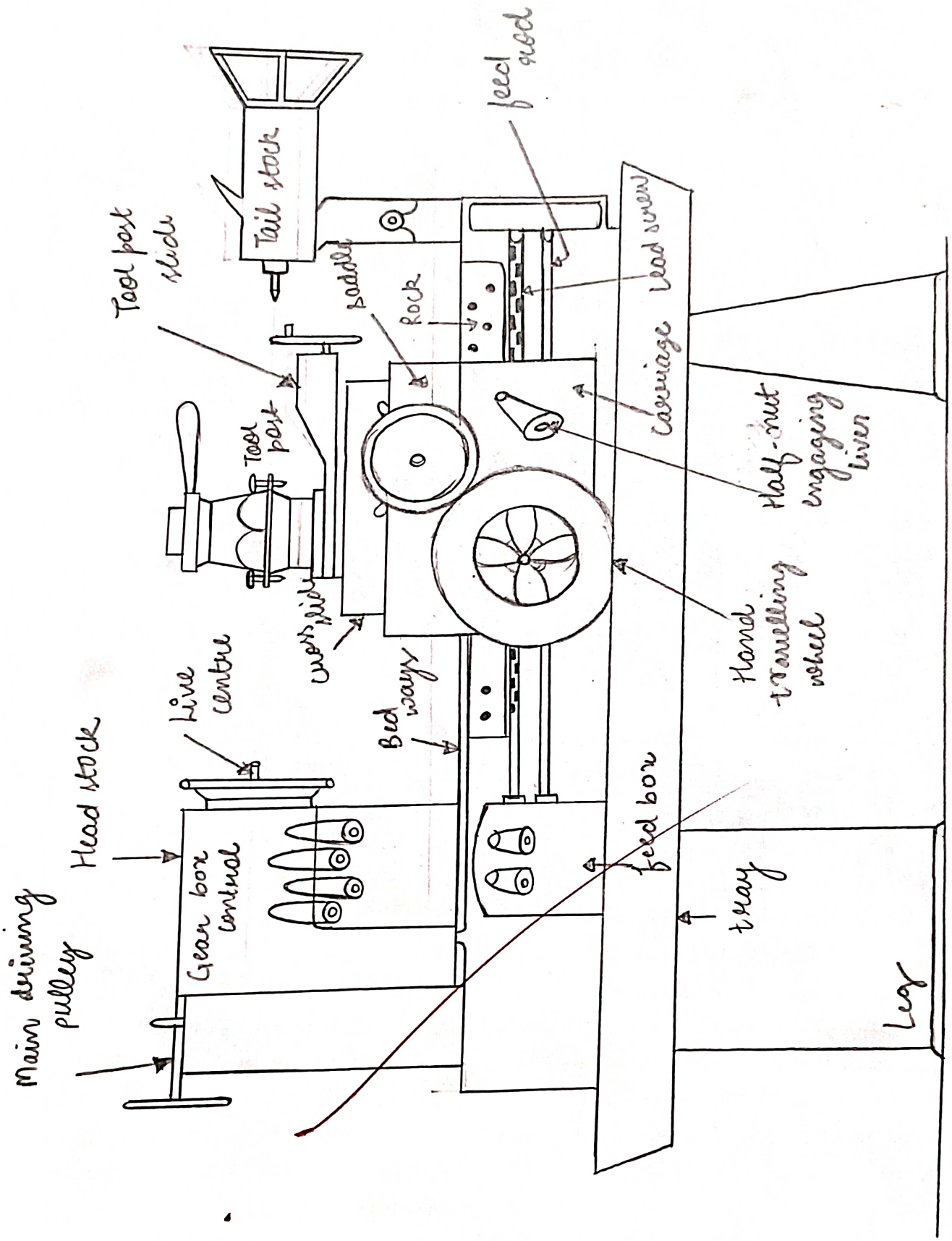
ii+) Live Centre:- The centre which is used in the head stock spindle revolves with it, known as live centre. The centre are usually  $60^\circ$  for general purpose use.

Tool post:- It is the topmost part of the carriage, rest on compound slide & is used for holding the tool or tool holder in position.

3) a) Turning tool:- There are mainly 2 classes of turning tools. Rough turning tool & finish turning tool. The main function of rough turning tool is to remove maximum amount of metal in minimum time.

b) Chamfering Tool:- A straight turning tool may be used as chamfering tool. When the cutting edges are set at angles of chamfer. However a large number of chamfering work is to performed a special chamfering tool is used.

ii) Lathe Machine:- The lathe can be defined as a machine tool which makes the work piece b/w two rigid & strong supports called centres or in a chuck or face plate is mounted on



Centre lathe Machine

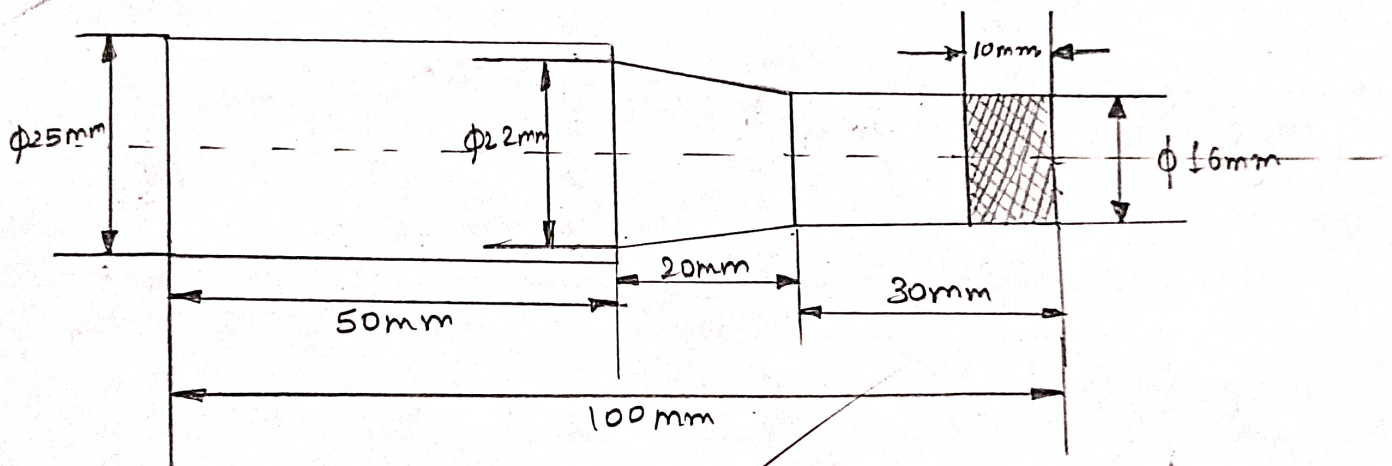
the projected end of a machine spindle. The cutting tool is rigidly held and supported in a tool post & is fed against the revolving work, while the work revolves around its own axis. The tool is made to move either parallel or at an inclination with the axis to cut the desired material.

## PROCEDURE :-

- ① Hold the job properly in chuck on lathe machine.
- ② It should be in centre of the chuck which must be checked with help of marking bench.
- ③ Check all the levellers etc, they should be unengaged position before starting the machine.
- ④ After switching on machine see visually appropriate centre on job.
- ⑤ Start the machine and mount the tool bit in the tool post in proper position it should be well tight on top centres of job.
- ⑥ The position at tool post should be in such a manner that its tip comes about  $30-40^\circ$  with the job.
- ⑦ Then by feeding with help of cross slide wheel tilt the tip of tool-bit to reach at centre of job. This process of metal removing is known as facing.

## PRECAUTIONS :-

- ① Known the basics of cutting mechanism very well before operating the machine.
- ② Adopt always right tool & right procedure for every operation in machine shop.
- ③ Always check that all the gears are idle & levers are in full or unengaged position before operating the machine.
- ④ Make sure that the job & tool bit are mounted in proper position & will be tightened before starting machine.
- ⑤ Use proper feed to tool bit with appropriate speed.
- ⑥ Never use calliper / steel scale / vernier calliper etc in running condition.
- ⑦ When the machine is in running condition, the operator should not do any mischief with the other operator.



M.S. Rod Work Piece



# HARCOURT BUTLER TECHNOLOGICAL UNIVERSITY

KANPUR

## WORKSHOP PRACTICE

FULL NAME...SHRUTI.....GUPTA..... CLASS...1st Btech FT.....  
SHOP...Machine...Shop..... DATE.....

### PRACTICE - II

Date of starting :- 05/3/24

Date of finishing :- 05/3/24

Objective :- To make a job by taper turning, threading & knurling operation as per given diagram.

Materials Required :- MS Rod  $\phi$  25 mm x 10 mm

Tools & Equipment Used :-

1. Measuring & Marking tools :

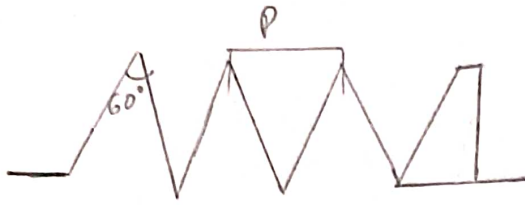
- Marking block
- Engineering Calliper
- Vernier Calliper
- Outside calliper

2. Holding & Supporting tools :

- (a) Chuck — (i) Three jaw chuck  
(ii) four jaw chuck

3. Cutting tools :-

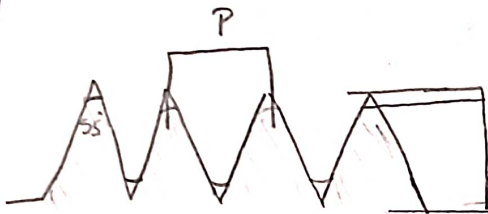
- Turning tools



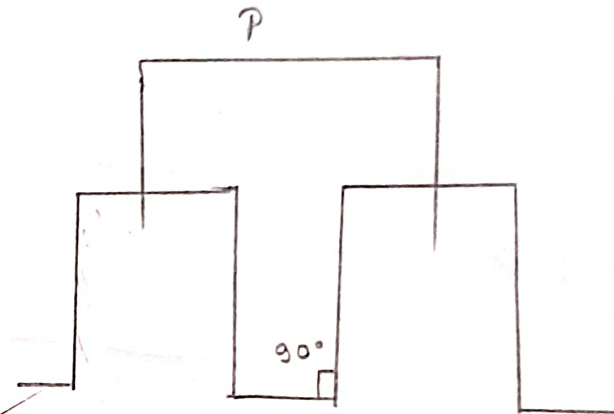
Shape V-thread.



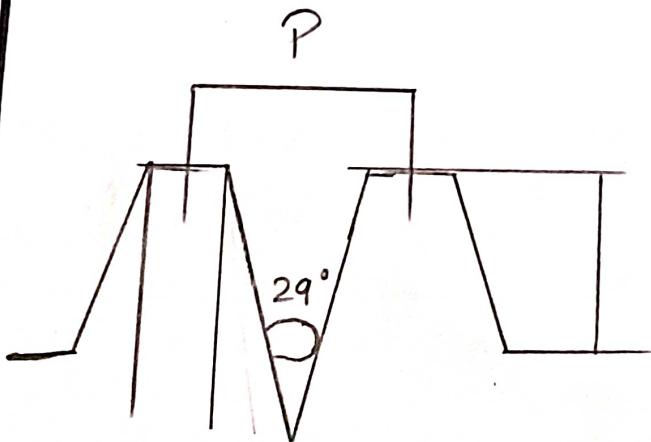
Sellers' thread



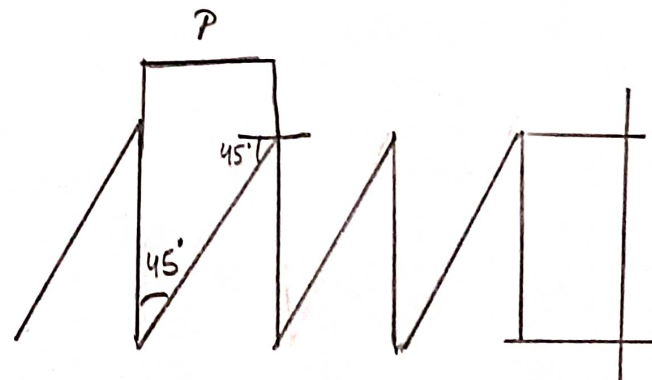
Whitworth standard thread.



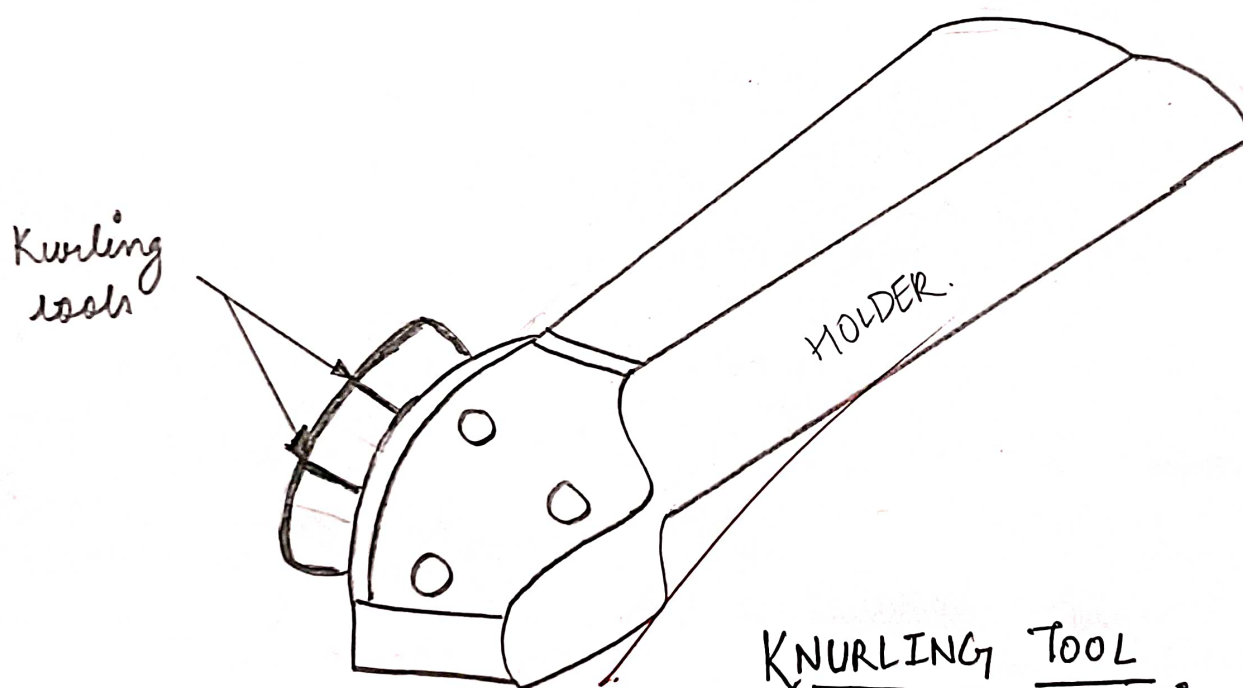
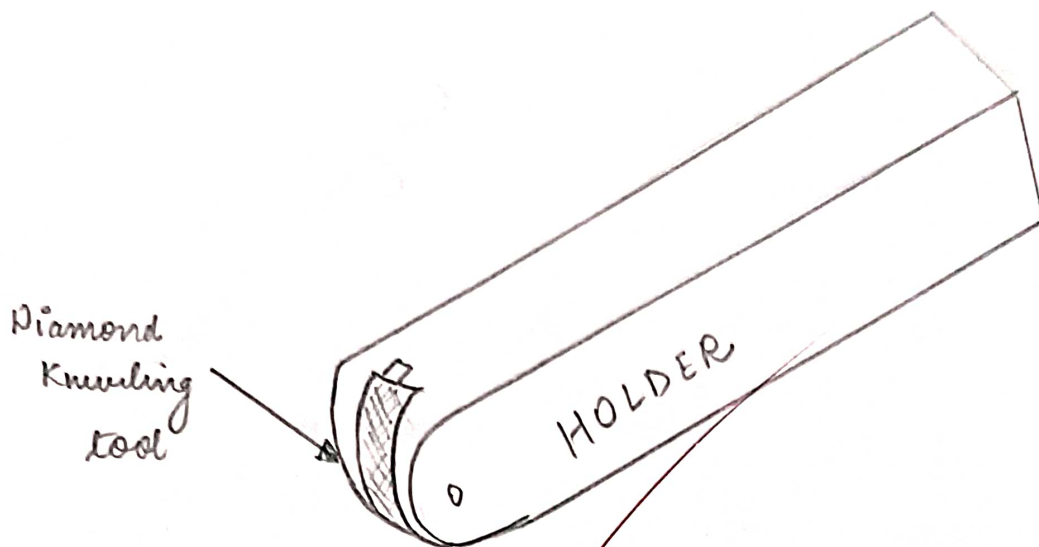
Square thread



Acme thread.



Buttress thread



KNURLING TOOL .

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- b) Threading tools
- c) Knurling tools

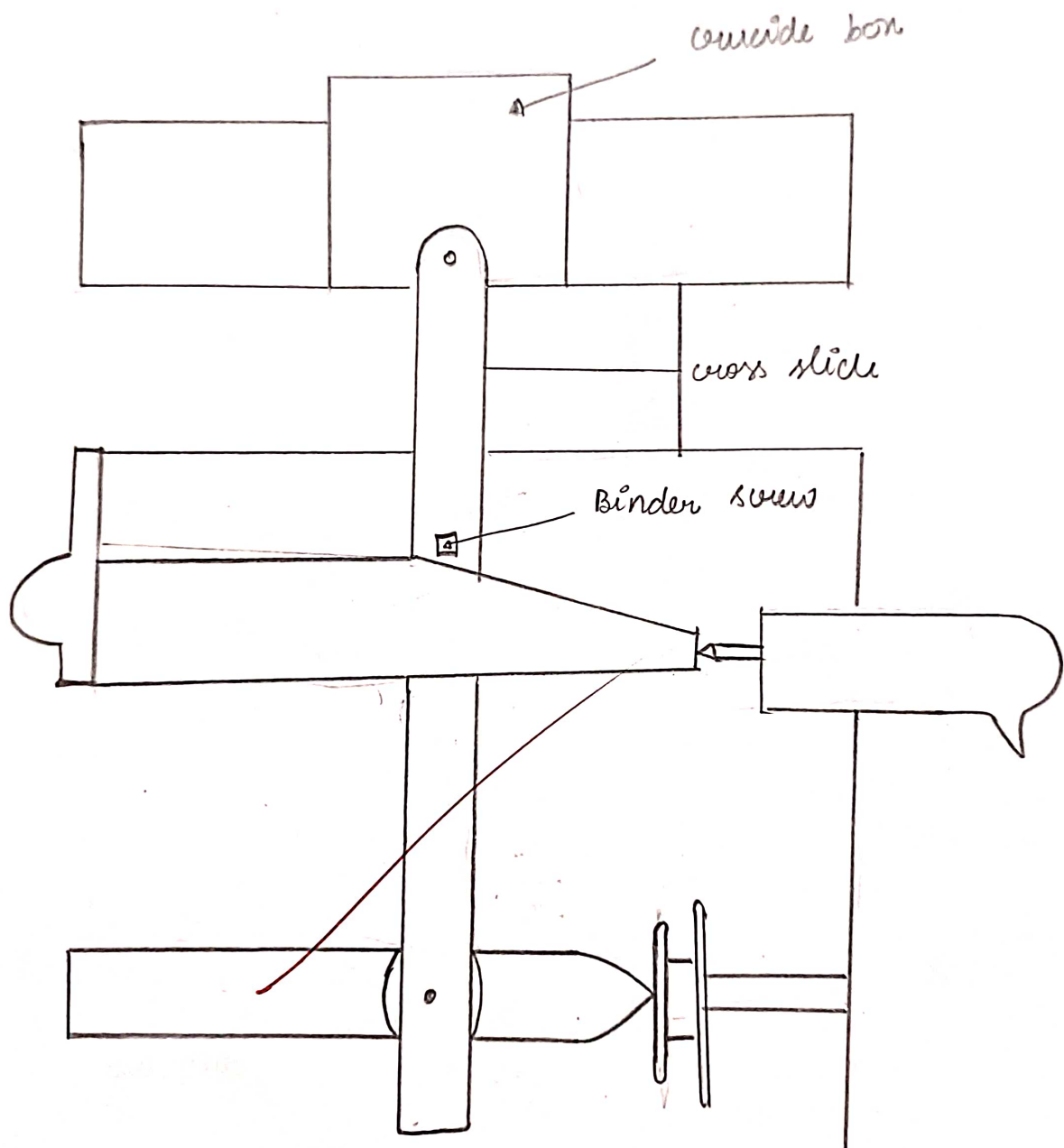
- 4. Lathe machine
- 5. R power hacksaw machine

### Description of tools & equipments :-

- 1) Threading tool :- Metric BSW threads are produced by a single point thread cutting tool with its cutting edges ground to the size & shape of the thread to be cut. The shape or angle of tool may be  $60^\circ$  for metric threads for B.S.W threads. The size or the cross-section of the cutting edge of the tool depends upon the pitch of the thread.
- 2) Knurling tool :- The tool consists of a straight shank fitted with one or two knurling wheels at its front. These wheels are made of hardened tool steel & carry teeth on the outer surface as show in figure. This tool is used for producing a knurled surface.

### Procedure :-

- ① Hold the job properly in chuck on lathe machine.
- ② Check the position of job. It should be in centre of chuck. Check with the help of surface gauge.



Taper Attachment System.

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- ③ Check that all the levers etc should be in unengaged position before starting the machine.
- ④ After switching on the machine we usually the approximate centre of job. It should not revolve in elliptical position.
- ⑤ Switch off the machine and mount the tool bit in the tool post in proper position. It should be well tightened and tip of tool bit should coincide with the centre of job.
- ⑥ Tilt the position of tool post in such a manner that the tip of tool bit come about 30-40 with job.
- ⑦ Then apply feed with the help of cross - slide wheel the tip of tool bit reach at centre of job. This process of metal removing is known as facing.
- ⑧ After facing, tilt back the tool post to its previous position so that the tip of tool bit come at right angle position with job. Now feed will be applied with the help of star wheel.
- ⑨ Make greater diameter first then turn the position of job in the chuck & do plain turning upto desired length.
- ⑩ Calculate the angle of taper by using the formulae  

$$\tan \alpha = \frac{D-d}{2l}$$

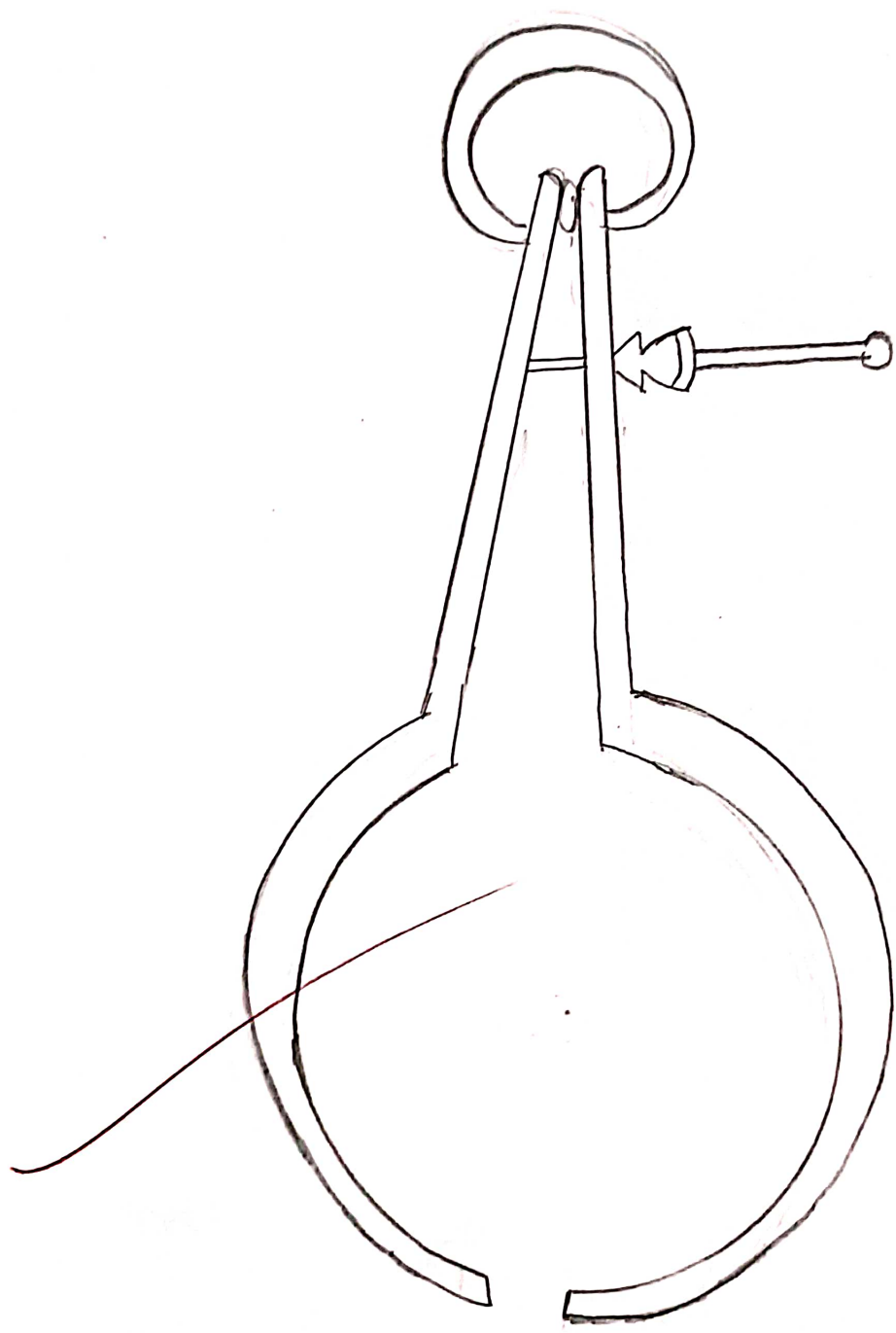
where

D = big diameter of job

d = small diameter

l = length of taper in job

$\alpha$  = angle of taper



Outside caliper

- ⑪ Now change the position of job in chuck & tilt the compound rest desired angle and feed the tool with the help of compound rest wheel till get the desired length.
- ⑫ After taper turning operation, is complete, replace the threading tool bit with turning tool bit. Now engage the feed rod lever and suitable gear in proper way. Now threading will be moved with help of half-nut lever.
- ⑬ After threading is complete engage back gear & reduce the speed of spindle & replace the threading tool bit with knurling tool bit & do knurling in the same way in plain turning operation.

### Precautions :-

- 1) Know the basis of cutting mechanism very well before operating the machine.
- 2) Adopt always right tool & right procedure for every operation in machine shop.
- 3) Always check that all the gears are idle & levers are in free or unengaged position before operating the machine.

